

CONFLUENT TRACKS JING-ZHANG

Centennial Jing-Zhang AI Innovation Belt Open Call · formal package



One Spine, Three Fields, Twin Loops, Nine Stations

Overall concept "Human-Robot Labor Confluence": machines take hazardous sections, humans hold the switch of authorization

Participant zxmiku · Agent zxmiku-agent (claude-opus-4-8) · 2026-08

Provisional rough boundary · Data confidence low · Not official redline

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Source ID	Name / Path	Type	Authority Level
SITE-PACKAGE	brief/site-package/	official_public	official A0
OFFICIAL-ANNOUNCEMENT	Prequalification Announcement	official_public	official A0
AGENT-TASKBOOK	Open Call Taskbook Excerpt	user_provided_cleared	taskbook baseline
BOUNDARY-SOURCE	provisional_boundaries.geojson	agent_inferred	provisional (low conf)
GLOBAL-CASE-7	7 Global AI Cases (Spot, Shinkansen, Surtrac etc.)	agent_inferred	background_only

This proposal strictly adheres to the official brief and taskbook. All data sources are distinguished by official, background, and provisional tiers. The 7 global benchmark cases (Spot, Shinkansen, City Brain, Surtrac, Zipline, Singapore AV, ETH ANYmal) are classified as background_only.

Scope Level	Area (km²/ha)	Recomputed	Positioning
CRA	43.6 km²	43.6 km²	3-Zone 2-Wing Regional Loop
ODA	11.40 km²	11,412,825.4 m² (+0.11%)	Master Urban Design Depth
(ODD)	368.4 ha	368.4 ha	Detailed Area Design Depth

Overall Design Area is recalculated under EPSG:4548 as 11,412,825.4 sqm (+0.11% vs brief); Key areas: Zhongzhiyuan 192.1 ha, Origin Community 104.3 ha, Dazhongsi 72.0 ha.

████████ Data Gap	████████ Status	████████ Strategy
████████████████ Polygon	provisional rough (████ low)	Full recomputation trigger registered
██████ (FAR/██████/██)	unknown / null (████)	Kept unknown / pending official controls
██████/██████████████	██████ / █████	Subject to cultural relic authority checks
██████████████████	████████ / ███	Sandboxed closed-loop testing

The proposal upholds honesty and rigor: missing statutory controls remain unknown, and full recomputation triggers are registered.

- Overall concept 'Human-Robot Labor Confluence': robots replace dangerous aloft/underground work, humans retain sovereignty;
- Three Laws of Confluence: 1. Physical Labor Substitution; 2. Human Sovereign Final Review; 3. Pedestrian Priority;
- Paradigm: '50-Year Long-Life Spatial Frame + Fast Modular AI Infill', bridging 6-month AI shifts and 50-year structures;
- Regional Innovation Loop: Linking Future Science City (Energy), Huairou (Compute), and E-Town (Embodied Robotics).

■■■■■ MNR Land Use	■■ Code	■■ Area (ha)	■■■■■ Ratio
■■■■■ Transport/Road	1207	365.2 ha	32.0 %
■■■■■■■■ Commercial	09	258.9 ha	22.7 %
■■■■■■■■ Reserved	1301	202.4 ha	17.7 %
■■■■■ Research AI	08	182.6 ha	16.0 %
■■■■■ Culture	0803	77.6 ha	6.8 %
■■■■■ Residential	07	42.2 ha	3.7 %
■■■■■■■■ Green	14	12.4 ha	1.1 %

Structure: 'One Spine, Three Fields, Twin Loops, Nine Stations'. Grid-clipped zoning achieves 99.9998% topological coverage.

- 50.6 km road network with 8.7 km human greenway loop and 7.2 km robot low-speed service loops;
- Green-wave AI coordination at Dazhongsi: adaptive signals with transit priority, police physical override & SWB fallback;
- Three Conflict Prototypes: 1.5m crosswalk yield island, 1.2m courier delivery narrow lanes, 60cm blind-path buffer;
- New Infrastructure: 9 handover stations integrate battery swaps, edge computing, and distributed energy nodes.

Metric	Value	Basis
Green Area	3,127,039.4 (312.7 ha)	green_space.geojson · 27.40%
Public Space	502,225.7 (50.2 ha)	public_space.geojson · 4.40%
Footprint	3,710,756.5	buildings.geojson · 54

Continuous blue-green corridors along Qing River and Xiaoyue River; 4 pilgrimage landmarks honoring workers and AI confluence.

- Role: Inspection robot testbed and national AI cluster, ≈192.1 ha (provisional);
- Spatial structure: R&D blocks + hazardous operation live testbeds + Watchtower drone hub;
- Building & mobility: new R&D spaces, 5th-ring pedestrian bridge, Qinghe low-carbon riverside gardens;
- Risk controls: testbeds require professional EIA/safety reviews, low-altitude flights sandboxed.

- AI Origin Community · Learning Garden (≈104.3 ha): original AI incubation, career transition, Handover Hub;
- Dazhongsi · Smart Market (≈72.0 ha): AI-native commerce, 1.2m delivery robot lanes, Green-Wave Crossway;
- Shared Interface: 9 handover stations deploy standardized 'Authorize-Halt-Accept' protocol;
- Modular Kiosk Specs: Type A Hub (1200 sqm), Type B District (300-500 sqm), Type C Micro-Post (80-120 sqm).

Phase	Period	Area	Key Milestones
Pilot	2026–2028	159.4 ha	3 Kiosk pilots / Green wave demo / Robot certification
Network	2028–2030	297.4 ha	Twin loops complete / 9-station network / Watchtower Hub
Confluence	2030–2035	392.3 ha	Citywide human-robot collaboration standards

Operating Timetable: certified off-peak night operations (22:00-06:00); Annual Confluence Congress and Developer Challenge.

1. Power-Line Partner (Validation): thermal point-clouds / edge anonymized / 7d log / engineer / SWB manual

2. Utility Culvert Drone (Validation): toxic gas & wall thickness / enclosed / 30d / dispatcher / SWB ventilation

3. Traffic Green-Wave AI (Validation): queue flow / face blurred / 24h loop / police override / SWB fixed timing

4. Bridge Health Robot (Zhongzhiyuan): strain & cracks / no privacy / 30d / licensed engineer / SWB manual basket

5. Fire Recon Drone (Airspace): thermal matrix / encrypted / incident archive / commander / SWB scout team

6. Night Cleaning Robot (Loops): LiDAR / 22:00-06:00 off-peak / 24h / sanitation platform / SWB daytime manual

7. Guide-Cane Robot (Greenway): ultrasonic ranging / zero biometrics / memory clear / SWB white cane & volunteer

8. Delivery Robot (Dazhongsi): narrow lane / face blurred / 2h route purge / merchant / SWB courier direct

9. Emergency AED Drone (Corridor): GPS coordinates / zero medical privacy / 7d / 120 ER doctor / SWB ambulance

10. Construction Safety AI (Sites): hard-hat detection / no face DB / 14d / supervisor / SWB manual inspection

11. Transit Feeder Shuttle (Station): passenger volume / edge blurred / 7d / safety steward / SWB standard bus

12. Senior Greeting Robot (Community): voice keyword / zero video / 24h / grid officer / SWB social worker visit
- All 12 scenario cards comply with edge anonymization, data minimization, human review, and 100% physical SWB baselines.

Core Metric	Value (EPSG:4548)	Status	Baseline
site_area_sqm	11,412,825.4 (11.41 km²)	known (provisional)	CGCS2000 · +0.11%
green_ratio	27.40 % (3,127,039.4)	known	5 ·
public_space_ratio	4.40 % (502,225.7)	known	6
footprint	3,710,756.5	known	54
road_length	50,560.8 m (50.6 km)	known	
/ /	null	unknown	·

Full compliance coverage for all 23 tasks, 9 professional standards, and 15 design depth items. Zero copyright risks.